

Notation

$\bar{w} = h/R$	true height of the particle centre, units of R
$\bar{w}_d, \Delta\bar{w}_d[k] = \bar{w}_d[k+1] - \bar{w}_d[k]$	commanded height and commanded step
$\delta\bar{w}_i$	tracking deviation, $\bar{w} = \bar{w}_d - \delta\bar{w}_3$; $\delta\bar{w}_1 =$ the same two steps later ($d=2$)
$\bar{a} = a/a_{\text{nom}}$	gain, normalised by the far-field gain
$\bar{a}' = d\bar{a}/d\bar{w}, \bar{a}'' = d^2\bar{a}/d\bar{w}^2$	slope and curvature of the gain in height
b	the law's constant; $b_{\text{true}}(\bar{w})$ the exact factor of the true curve
$A = \partial\bar{a}'/\partial\bar{a}$	self-sensitivity of the slope
$\hat{\theta}$	estimator's value of θ ; $\bar{a}', \bar{a}'', A$ without subscript are evaluated at $(\hat{a}_w, \hat{b})$
$e_\theta = \theta - \hat{\theta}$	error, TRUE minus ESTIMATOR (SSOT convention)
$\Delta\bar{w}, \Delta\hat{\bar{w}}, e_{\Delta\bar{w}} = \Delta\bar{w} - \Delta\hat{\bar{w}}$	true step, estimator's step, step error
F_{dw}	loop coupling: the normalised force the controller applied (previous step)
ε_w	thermal step of the position (variance Q_{33})
n_w, n_a	position and gain-readout measurement noise (R_1, R_2)
λ_c	pole of the position loop
$\nabla_d\bar{w}_d$	commanded displacement the delayed readout walks back
$P_{ij} = \mathbb{E}[e_i e_j]$	error covariance; F_e, H, Q its propagation

1 Gain model

Law, its derivatives and its solution (the estimator's model; $\hat{b}$ constant)

$$\begin{aligned}
\bar{a}' &= b(1 - \bar{a})^2 \\
A &= \frac{\partial\bar{a}'}{\partial\bar{a}} = -2b(1 - \bar{a}) \\
\frac{\partial\bar{a}'}{\partial b} &= (1 - \bar{a})^2 = \frac{\bar{a}'}{b} \\
\bar{a}'' &= \frac{d\bar{a}'}{d\bar{w}} = A\bar{a}' = -2b(1 - \bar{a})\bar{a}' \\
\frac{1}{1 - \bar{a}(\bar{w})} &= b(\bar{w} - \bar{w}_s) \quad \Longleftrightarrow \quad \bar{a}(\bar{w}) = 1 - \frac{1}{b(\bar{w} - \bar{w}_s)}
\end{aligned}$$

True curve (published plane: $b_{\text{true}}(\bar{w})$ a function, $0.883 \rightarrow 0.867$ at $\bar{w} = 2.19 \rightarrow 0.928$ at $\bar{w} = 1.10$)

$$\begin{aligned}
\bar{a}'_{\text{true}}(\bar{w}) &= b_{\text{true}}(\bar{w}) (1 - \bar{a}_{\text{true}}(\bar{w}))^2 \\
\bar{a}''_{\text{true}}(\bar{w}) &= b'_{\text{true}}(\bar{w}) (1 - \bar{a}_{\text{true}})^2 - 2b_{\text{true}}(\bar{w}) (1 - \bar{a}_{\text{true}}) \bar{a}'_{\text{true}}
\end{aligned}$$

Model error of a constant $\hat{b}$ (the only approximation of the law), and the true slope and curvature written around the estimator's point

$$\begin{aligned}
e_b[k] &:= b_{\text{true}}(\bar{w}[k]) - \hat{b}[k] \\
f[k] &:= (1 - \hat{a}_w[k])^2 e_b[k] \\
\bar{a}'_{\text{true}}(\bar{w}[k]) &= \bar{a}' + A e_{\bar{a}_w}[k] + f[k] + \hat{b} e_{\bar{a}_w}[k]^2 - 2(1 - \hat{a}_w) e_{\bar{a}_w}[k] e_b[k] \\
\bar{a}''_{\text{true}}(\bar{w}[k]) &= \bar{a}'' + f'[k] + \frac{\bar{a}''}{\bar{a}'} f[k] + \mathcal{O}(e), \quad f' = \frac{df}{d\bar{w}}
\end{aligned}$$

2 True dynamics

$$\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k] \\
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k] \\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c \delta\bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\Delta\bar{w}[k] &= \Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
\bar{a}_w[k+1] &= \bar{a}_w[k] + \bar{a}'_{\text{true}}(\bar{w}[k]) \Delta\bar{w}[k] + \frac{1}{2} \bar{a}''_{\text{true}}(\bar{w}[k]) \Delta\bar{w}[k]^2 + \mathcal{O}(\Delta\bar{w}^3) \\
b_{\text{true}}(\bar{w}[k+1]) &= b_{\text{true}}(\bar{w}[k]) + b'_{\text{true}}(\bar{w}[k]) \Delta\bar{w}[k] + \mathcal{O}(\Delta\bar{w}^2) \\
y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k] - \bar{a}'_{\text{true}}(\bar{w}[k]) \nabla_d \bar{w}_d[k] + n_a[k]
\end{aligned}$$

3 Estimator

State $\hat{x} = [\delta\hat{w}_1, \delta\hat{w}_2, \delta\hat{w}_3, \hat{a}_w, \hat{b}]^\top$; predict

$$\begin{aligned}
\delta\hat{w}_1[k+1] &= \delta\hat{w}_2[k] \\
\delta\hat{w}_2[k+1] &= \delta\hat{w}_3[k] \\
\delta\hat{w}_3[k+1] &= \lambda_c \delta\hat{w}_3[k] \\
\Delta\hat{w}[k] &= \Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k] \\
\hat{a}_w[k+1] &= \hat{a}_w[k] + \bar{a}' \Delta\hat{w}[k] + \frac{1}{2} \bar{a}'' \Delta\hat{w}[k]^2 + m_2[k] \\
\hat{b}[k+1] &= \hat{b}[k]
\end{aligned}$$

($m_2[k]$ = the mean of the second-order error group, Section 4, added back so that $\mathbb{E}[e_{\bar{a}_w}]$ has no drift; the exact-step form $1/(1 - \hat{a}_w[k+1]) = 1/(1 - \hat{a}_w[k]) + \hat{b} \Delta\hat{w}[k]$ replaces the two law terms in the code and agrees with them to second order)

Predicted measurements

$$\begin{aligned}
\hat{y}_1[k] &= \delta\hat{w}_1[k] \\
\hat{y}_2[k] &= \hat{a}_w[k] - \bar{a}' \nabla_d \bar{w}_d[k]
\end{aligned}$$

4 Error dynamics

Step error and its moments ($u := e_{\Delta\bar{w}}$)

$$\begin{aligned}
e_{\Delta\bar{w}}[k] &= \Delta\bar{w}[k] - \Delta\hat{w}[k] = (1 - \lambda_c) e_{\delta\bar{w}_3}[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
\mathbb{E}[e_{\Delta\bar{w}}] &= 0, \quad \text{Var}(e_{\Delta\bar{w}}) = (1 - \lambda_c)^2 P_{33} + F_{dw}^2 P_{44} + 2(1 - \lambda_c) F_{dw} P_{34} + Q_{33} \\
\text{Cov}(e_{\delta\bar{w}_3}, e_{\Delta\bar{w}}) &= (1 - \lambda_c) P_{33} + F_{dw} P_{34} \\
\text{Cov}(e_{\bar{a}_w}, e_{\Delta\bar{w}}) &= (1 - \lambda_c) P_{34} + F_{dw} P_{44} \\
\text{Cov}(e_b, e_{\Delta\bar{w}}) &= (1 - \lambda_c) P_{35} + F_{dw} P_{45}
\end{aligned}$$

Position chain (exact)

$$e_{\delta\bar{w}_1}[k+1] = e_{\delta\bar{w}_2}[k]$$

$$e_{\delta\bar{w}_2}[k+1] = e_{\delta\bar{w}_3}[k]$$

$$e_{\delta\bar{w}_3}[k+1] = \lambda_c e_{\delta\bar{w}_3}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k]$$

Gain, all terms to second order (true increment minus estimator increment, $\Delta\bar{w} = \Delta\hat{w} + e_{\Delta\bar{w}}$)

$$\begin{aligned} e_{\bar{a}_w}[k+1] - e_{\bar{a}_w}[k] &= \left[\bar{a}' + A e_{\bar{a}_w} + f + \hat{b} e_{\bar{a}_w}^2 - 2(1 - \hat{a}_w) e_{\bar{a}_w} e_b \right] (\Delta\hat{w} + e_{\Delta\bar{w}}) \\ &\quad + \frac{1}{2} \left[\bar{a}'' + f' + \frac{\bar{a}''}{\bar{a}'} f \right] (\Delta\hat{w} + e_{\Delta\bar{w}})^2 - \bar{a}' \Delta\hat{w} - \frac{1}{2} \bar{a}'' \Delta\hat{w}^2 - m_2 \\ &= \underbrace{\bar{a}' e_{\Delta\bar{w}} + A \Delta\hat{w} e_{\bar{a}_w} + \Delta\hat{w} f}_{(1) \text{ first order}} + \underbrace{A e_{\bar{a}_w} e_{\Delta\bar{w}} + f e_{\Delta\bar{w}} + \bar{a}'' \Delta\hat{w} e_{\Delta\bar{w}} + \frac{1}{2} \bar{a}'' e_{\Delta\bar{w}}^2}_{(2) \text{ second order, the } \bar{a}'' \text{ group}} \\ &\quad + \underbrace{\frac{1}{2} \left(f' + \frac{\bar{a}''}{\bar{a}'} f \right) (\Delta\hat{w} + e_{\Delta\bar{w}})^2 + \hat{b} e_{\bar{a}_w}^2 (\Delta\hat{w} + e_{\Delta\bar{w}}) - 2(1 - \hat{a}_w) e_{\bar{a}_w} e_b (\Delta\hat{w} + e_{\Delta\bar{w}}) - m_2}_{(3) \text{ second order, model-error curvature and squares}} + \mathcal{O}(3) \end{aligned}$$

Law constant (the constant model has no term for the second line)

$$e_b[k+1] = e_b[k] + b'_{\text{true}}(\bar{w}[k]) \Delta\bar{w}[k]$$

Mean of the second-order group (this is m_2 ; $\mathbb{E}[e] = 0$ for every state, $\mathbb{E}[e_{\Delta\bar{w}}] = 0$)

$$\begin{aligned} m_2[k] &= A \text{Cov}(e_{\bar{a}_w}, e_{\Delta\bar{w}}) + \frac{1}{2} \bar{a}'' \text{Var}(e_{\Delta\bar{w}}) + \frac{\bar{a}'}{\hat{b}} \text{Cov}(e_b, e_{\Delta\bar{w}}) \\ &\quad + \frac{1}{2} \left(f' + \frac{\bar{a}''}{\bar{a}'} f \right) (\Delta\hat{w}^2 + \text{Var}(e_{\Delta\bar{w}})) + \hat{b} P_{44} \Delta\hat{w} - 2(1 - \hat{a}_w) P_{45} \Delta\hat{w} \end{aligned}$$

(code: `pred_mean2` carries the first two terms, `pred_mean2_e4` the $A \text{Cov}(e_{\bar{a}_w}, e_{\Delta\bar{w}})$ form and the e_b line; the f, f' terms need b_{true} and are NOT in the estimator; the two square terms are $\mathcal{O}(P) \Delta\hat{w}$ and are dropped)

First-order part written as $e[k+1] = F_e e[k] + g \varepsilon_w[k] + (\text{model error})$, $e = [e_{\delta\bar{w}_1}, e_{\delta\bar{w}_2}, e_{\delta\bar{w}_3}, e_{\bar{a}_w}, e_b]^\top$

$$F_e = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 \\ 0 & 0 & (1 - \lambda_c) \bar{a}' & 1 + \bar{a}' F_{dw} + A \Delta\hat{w} & \frac{\bar{a}'}{\hat{b}} \Delta\hat{w} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad g = \begin{bmatrix} 0 \\ 0 \\ -1 \\ \bar{a}' \\ 0 \end{bmatrix}, \quad Q = Q_{33} g g^\top$$

$$\text{row 4: } \Delta\hat{w}[k] f[k] = \Delta\hat{w}[k] (1 - \hat{a}_w[k])^2 e_b[k] = F_e(4, 5) e_b[k]$$

$$\text{row 5: } b'_{\text{true}}(\bar{w}[k]) \Delta\bar{w}[k]$$

(row 4: the product is already in F_e , but $\mathbb{E}[e_b] \neq 0$ – a curve minus a constant – so it carries a MEAN;
row 5: no container in F_e or Q)

Measurement Jacobians ($\partial y / \partial x$ at the estimate)

$$H_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$H_2 = \begin{bmatrix} 0 & 0 & 0 & 1 - A \nabla_d \bar{w}_d & -\frac{\bar{a}'}{\hat{b}} \nabla_d \bar{w}_d \end{bmatrix}, \quad R = \text{diag}(R_1, R_2)$$

Where each entry comes from

$$\begin{aligned}
 F_e(3, 4) = -F_{dw} : & \quad \text{force computed with } \hat{\bar{a}}_w, \text{ position moved by } \bar{a}_w \\
 F_e(4, 3) = (1 - \lambda_c) \bar{a}' : & \quad \partial \Delta \hat{w} / \partial \delta \hat{w}_3 = 1 - \lambda_c \\
 F_e(4, 4) = 1 + \bar{a}' F_{dw} + A \Delta \hat{w} : & \quad \bar{a}' \partial e_{\Delta \bar{w}} / \partial e_{\bar{a}_w} = \bar{a}' F_{dw}; \quad A \Delta \hat{w} = \text{the slope reads } \hat{\bar{a}}_w \\
 F_e(4, 5) = (\bar{a}' / \hat{b}) \Delta \hat{w} : & \quad \partial \bar{a}' / \partial \hat{b} \times \text{step}; 0 \text{ in a hold and at a turning point} \\
 H_2(4), H_2(5) : & \quad \text{the readout walks the slope back over } \nabla_d \bar{w}_d
 \end{aligned}$$